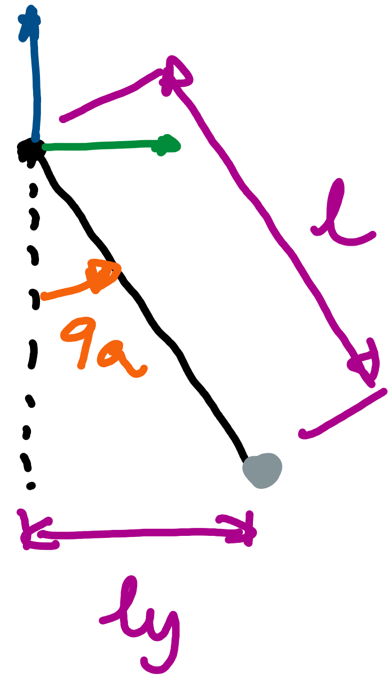
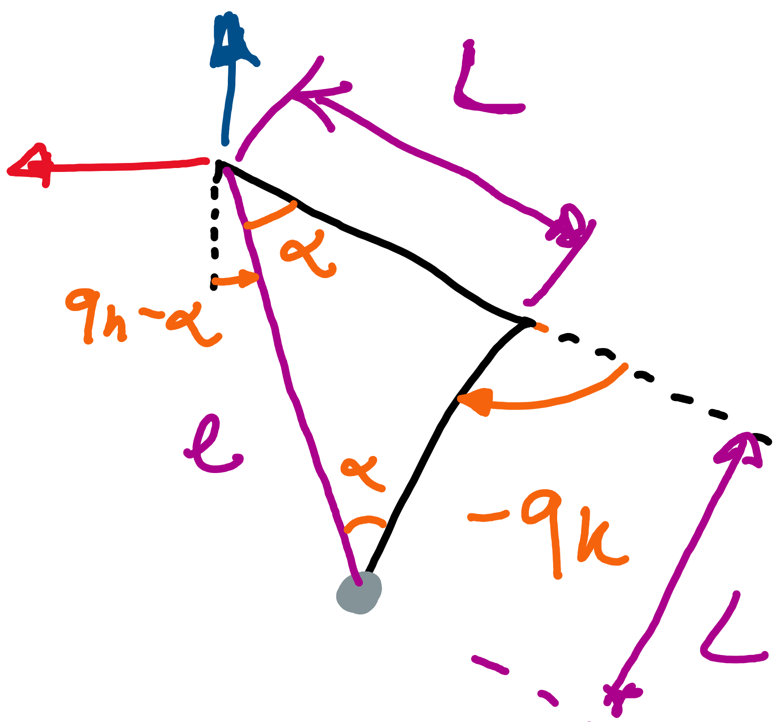
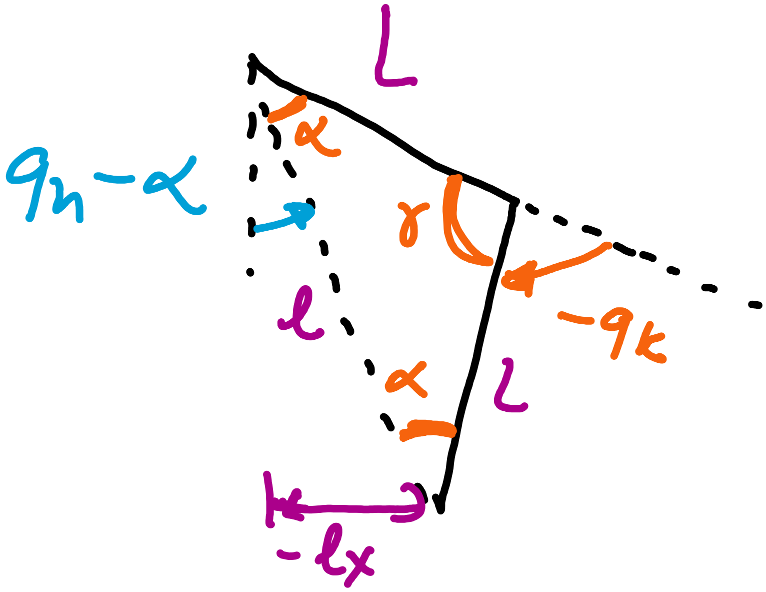

l
qa
ly


L
α
qh-α
l
α
-qk
L



Trignometry: $\gamma - q_k = \pi \Rightarrow q_k = -\pi + \gamma$

Cosine rule: $l^2 = 2L^2(1-\cos\gamma)$

$\gamma = \cos^{-1}\left[(2L^2 - l^2)/2L^2\right]$

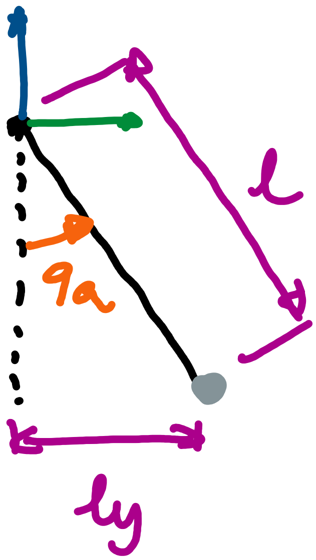
Solve for q_k: $\boxed{q_k = -\pi + \cos^{-1}\left[\frac{2L^2-l^2}{2L^2}\right]}$

Sum of angle in a triangle: $2\alpha + \gamma = \pi \Rightarrow 2\alpha + \cancel{\pi} + q_k = \cancel{\pi}$

$\alpha = -0.5\, q_k$

Trignometry $q_h - \alpha = \sin^{-1}\left(-\frac{l_x}{l}\right)$

$\boxed{q_h = -0.5\, q_k + \sin^{-1}\left(-\frac{l_x}{l}\right)}$



$$l = \sqrt{l_x^2 + l_y^2 + l_z^2}$$

$$\boxed{q_a = \sin^{-1}\left(\frac{l_y}{l}\right)}$$